

A counterexample to Conjecture 00000004396

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The conjecture asserts, in particular, that every Möbius ladder is Sidorenko. This assertion is false for the eight-vertex Möbius ladder. No restriction to bipartite ladders or to a congruence class of their orders appears in either language of the source.

For finite simple graphs H, G , write

$$t(H, G) = \frac{\text{hom}(H, G)}{|V(G)|^{|V(H)|}}.$$

Here a homomorphism is an arbitrary vertex map preserving every edge; it need not be injective. The Sidorenko property is

$$t(H, G) \geq t(K_2, G)^{|E(H)|} \quad \text{for every nonempty finite graph } G.$$

This is the standard finite-graph formulation[1].

Let H have vertex set $\mathbb{Z}/8\mathbb{Z}$, with edges between consecutive vertices and between opposite vertices. Thus H is the cycle C_8 together with the four edges $\{i, i+4\}$, $0 \leq i < 4$. It has twelve edges and is a Möbius ladder independently of conventions about including the smallest four-vertex member of this family.

The vertices $0, 1, 2, 3, 4, 0$ form a cycle of length five in H . A homomorphism $H \rightarrow K_2$ would alternate its two values along every edge of this cycle, which is impossible. Consequently

$$\text{hom}(H, K_2) = 0, \quad t(H, K_2) = 0.$$

On the other hand, K_2 has two ordered adjacent pairs out of four ordered vertex pairs, so

$$t(K_2, K_2) = \frac{1}{2}, \quad t(K_2, K_2)^{|E(H)|} = 2^{-12} > 0.$$

The required inequality is therefore false. This refutes the conjunction in the source; no claim about odd subdivisions of trees is needed.

Formal verification. The self-contained Lean 4.19.0 file `lean/Main.lean` defines actual finite simple graphs, edge-preserving vertex maps, and the Sidorenko inequality. Counts use complete duplicate-free lists of maps. The inequality is written after multiplication by positive integer denominators, so there is no approximation or division convention:

$$(2|E(G)|)^{|E(H)|} |V(G)|^{|V(H)|} \leq \text{hom}(H, G) |V(G)|^{2|E(H)|}.$$

The proof first excludes every map $\text{Fin } 8 \rightarrow \text{Fin } 2$ directly using the five-cycle. It then proves that the 256 binary codes enumerate all such functions, each exactly once, using inverse encoding and decoding maps. The computed counts are thus counts of the intended objects. The final theorem `conjecture_4396_false` negates the universal assertion about the cycle-plus-opposite-edge family. The file uses `no_sorry`, `admit`, `native_decide`, or additional axiom.

Previous withdrawn submission. An earlier, independently authored submission by `orionsheep` used the smaller K_4 member and the same necessary bipartiteness observation. It was withdrawn by its author, with no review rejection recorded, in pull request #318. The present package supplies an independent eight-vertex construction and full finite-model verification; it does not claim priority for the general bipartiteness obstruction.

References

- [1] *Sidorenko's conjecture, colorings and independent sets*, <https://arxiv.org/abs/1603.05888>.